



Nutritional Assessment (HuNu3081)

Chapter One: Introduction to Nutritional Assessment

Objectives

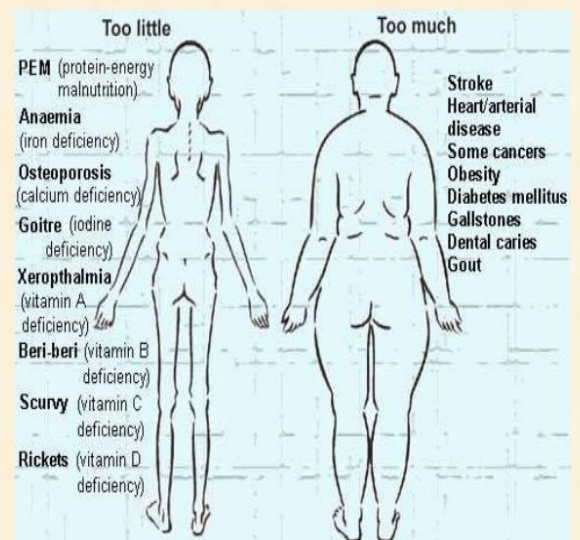
At the end of this chapter you will be able to:

- Define Nutritional Status and Nutritional Assessment.
- Classify Nutritional Assessment Methods.
- Discuss the purposes of Nutritional Assessment.
- Describe Nutritional Assessment Systems: Surveys, Surveillance, Screening and Intervention.
- Identify Nutritional Indices and Indicators.

1. Introduction

People need to maintain optimal nutritional status in order to support the body's growth and development, promote health, and prevent diseases. There is an important relationship between nutrition and health. Optimal nutritional status promotes health & wellbeing, prevent diseases and improve the quality of life. On the other hand sub optimal nutritional status can compromise immunity and leads to poor health status. Likewise, good health status may contribute for the achievement of optimal nutritional status and poor health status may compromise the nutritional status of individuals and leads to malnutrition.

Examples of illnesses caused by improper nutrient consumption



In 1674, Sydenham, a British physician recognized that *iron in wine can treat anemia*, and another British naval surgeon namely Lind in 1740s discovered that *consuming limes and lemons can cure scurvy in sailors*). In 1816, a German physician called Magendie found that *dogs fed only on cabs rapidly lost body weight and died in a few weeks*.

As a result; nutritional assessments need to be conducted to determine the **nature, magnitude and causes of malnutrition** and improve nutritional status and living conditions of communities. Nutritional assessment procedures were first used in surveys designed to describe the nutritional status of populations on a national basis. In some countries, data are collected on an ongoing basis using nutrition surveillance systems. In the past, these systems were targeted to high-risk populations, especially low-income mothers, children under five years, and pregnant women. Now with the growing awareness of the role of nutrition as a risk factor for chronic diseases, surveillance systems often encompass all age groups.

The information obtained from nutritional assessment can be used to determine the nutritional status of individuals or population groups, identify public health nutrition problems and design effective intervention programs.

Definition of Terms

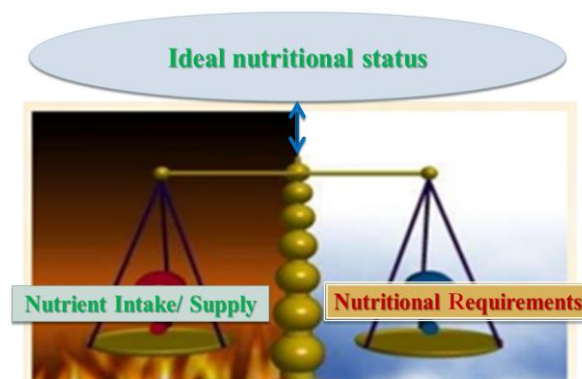
- ☑ **Nutrition** can be defined as the process of taking food into the body and absorbing the nutrients in those foods. It is the process of nourishing or being nourished; *specifically*: the sum of the processes by which an animal or plant takes in and utilizes food substances.
- ☑ **Nutritional status** is the current body status/ health status of a person or population groups related to their state of nourishment (*i.e.* the intake and utilization of nutrients).
- ☑ **Nutritional Assessment** is the process of gathering, analyzing and interpretation of information from Anthropometric, Biochemical/ Laboratory, Clinical and Dietary studies.

1.1. Nutritional Status

- It is often resulted from many inter-related factors including **diet/ food intake** (**quantity & quality**), **physical health** and **infections**. It can be determined by a complex interaction between;
 - ✓ Internal factors (**age, sex, behavior, physical activity, diseases**) and
 - ✓ External environmental factors (**food safety, cultural, social & economic circumstances**).

- Nutritional status ranges from **obesity** to **severe under nutrition**. Incidence of malnutrition is high among the low income groups in developing countries, particularly among pregnant & lactating women, weaned infants and preschool children.
- **When an ideal Nutritional Status could be achieved?**

An ideal nutritional status occurs when the **supply of nutrients** conform to the **nutritional requirements or needs** of individuals.



The nutritional status of an individual has consequences:

- + **Optimal nutritional status:** promotes health & wellbeing, prevent diseases and improve the quality of life.
- + **Malnutrition:** increases the risk of (susceptibility to) infections and chronic diseases.
- + **Under nutrition:** leads to infections and decreases in growth, physical & mental development in children and affect the health & physical efficiency of adults.
- + **Over nutrition:** leads to obesity and its comorbidities (non-communicable diseases (NCDs)).

1.2. What is Nutritional Assessment?

Nutritional Assessment involves collecting data about a client's medical history, anthropometric measurements, clinical and biochemical characteristics, dietary practices, current treatment, and socioeconomic situation including food security.

1.3. Why Nutritional Assessment?

Nutritional assessment is the first step in the treatment of malnutrition.

📖 **The purpose of nutritional assessment is to:**

- ☑ Identify individuals or population groups at risk of becoming malnourished.
- ☑ Identify individuals or population groups who are malnourished.
- ☑ Develop health care programs that meet the community needs which are defined by the assessment.
- ☑ Measure the effectiveness of the nutritional programs & interventions that are introduced.

1.4. When Nutritional Assessment is most critical?

Nutritional measurements are especially critical when nutritional risks are suspected (when there are **at-risk individuals**), during periods of **rapid growth & development**, when **nutritional requirements are high**, and inadequate nutrition can have **irreversible health consequences**.

1.5. Nutritional Assessment Methods

Nutritional assessment methods can be classified in to two broad categories; *i.e.* **direct** and **indirect** methods.

1. Direct methods: deal with the individual and measure objective criteria. **These are summarized as ABCD.**

- ☑ **A**nthropometric measurements,
- ☑ **B**iochemical methods/ Laboratory tests
- ☑ **C**linical evaluations &
- ☑ **D**ietary assessment methods

2. Indirect methods of nutritional assessment: use community health indices that reflect community nutritional status/ needs and their influences. **These include:**

- ☑ **Ecological variables** such as: agricultural crop production.
- ☑ **Economic factors:** *e.g.* household income, per capita income, population density, food availability & prices etc.
- ☑ **Cultural & social habits**
- ☑ **Vital health statistics:** mortality, morbidity & other health indicators (*e.g.* infant & under 5 mortality), utilization of maternal & child health care services, fertility indices and sanitary conditions etc.

Note

- Each method by itself does not provide a complete picture of a person's nutritional profile, so they should be used in combination to obtain a more complete assessment.
- Based on the assessment result;
 - Appropriate interventions should be planned and implemented or;
 - Referrals should be done for further examinations.

1.6. Nutritional Assessment Systems

In the past, nutritional assessment systems were focused on methods to characterize each stage in the development of a nutritional deficiency state (*i.e. Adequacy→ Negative balance→ Decline in tissue store→ Loss of function, sign & symptoms, organ failure→ Death.*) But they are now applied to define multiple levels of nutrient status and not just the level associated with a nutrient deficiency. Nutritional assessment systems can take one of the following four forms:

- ☑ **Surveys,**
- ☑ **Surveillance,**
- ☑ **Screening, or**
- ☑ **Intervention.**

1.6.1 Nutrition Surveys

Nutritional assessment procedures were first used in surveys designed to describe nutritional status on a national basis.

Survey involves collecting data directly from people **only once** to describe what is happening at the present moment. It is mostly carried out through cross-sectional design. Cross-sectional surveys can identify and describe geographic areas and/or population sub-groups “at risk” for chronic malnutrition and unlikely to identify acute malnutrition or provide information on the possible causes of malnutrition.

Surveys are used to establish baseline nutritional status of a population and formulate policies. It involves drawing a conclusion by taking only a representative sample of individuals from a given community covering all ages and both sexes in different socio-economic groups.

1.6.1.1 Steps in Nutrition Surveys

A nutrition survey takes place in 5 steps:

- ☒ Collection of available information on the nutritional situation and other demographic, socioeconomic and ecological data in the survey region;
- ☒ Planning and preparation of the nutrition survey;
- ☒ Implementation of the nutrition survey;
- ☒ Data processing, evaluation and analysis;
- ☒ Dissemination of the results of the nutrition survey and preparation for translation into action.

1.6.1.2 Types of Surveys

- ☑ **Rapid Appraisal:** obtaining information on the nutritional condition of the target community using qualitative methods. So, Anthropometric data (such as height & weight) are not recorded.
- ☑ **Rapid Assessment:** obtain information on the type of nutritional problems using quantitative methods *i.e.* Anthropometric data are measured.
- ☑ **Baseline Survey:** gathering information concerning the type and prevalence of nutritional problems.
- ☑ **Follow-up Survey:** assessing the impact of the project on the nutritional condition of a community.

1.6.2 Nutrition Surveillance

Nutrition Surveillance involves the continuous monitoring of the nutritional status of the same population groups *i.e.* data collected over time. In the past, surveillance systems were targeted to high-risk populations (low-income mothers, children <5 years, & pregnant women). Now, nutrition is considered as a risk factor for chronic diseases, so that surveillance systems often encompass all age groups. It can be used to formulate and initiate intervention measures and monitor the effect of government policies and evaluating the efficacy and effectiveness of nutrition intervention programs.

Surveillance can identify: **seasonal differences, both acute & chronic malnutrition and their possible causes**. For Example;

- ☑ **Acute (Short-term)** nutritional effects identified by surveillance includes:
 - High intake of sugary foods – **increased dental caries**.
 - High intake of salt – **increased blood pressure**.
- ☑ **Chronic (Long-term)** nutritional effects identified by surveillance includes:
 - Lack of fruit and vegetable intake, high & prolonged intake of alcohol – **colon cancer**.
 - Lack of breastfeeding – **breast cancer**.
 - Central obesity – **type 2 diabetes**.
 - Lack of dietary calcium – **osteoporosis**.

1.6.3 Nutrition Screening

Screening involves collection of data on the whole population regardless of setting, age, or targeted to sub-population or individuals considered to be at nutritional risk.

It is used for the identification of malnourished or “at risk” individuals requiring intervention. It involves a comparison of measurements of individuals with predetermined risk levels or “cutoff” points. It can be carried out using measurements that are simple and cheap, and that can be applied rapidly on a large scale and can also be conducted by individuals other than the registered dietitian.

1.6.4 Nutrition Intervention

It often targets population subgroups identified as “at-risk” during nutrition surveys or nutrition screening. Examples of nutrition interventions are:-

- ▶ Nutrition education
- ▶ Supplementation
- ▶ Fortification
- ▶ Dietary approaches: diversification, modification, biofortification etc.

1.7 Nutritional Assessment Indices & Indicators

Raw measurements such as weight & height alone have no meaning unless they are related with each other or to the age of an individual.

- **Indices (singular: index)** are constructed from two or more raw measurements or a measurement plus age.
- Indices are necessary for interpretation and grouping of measurements and are often evaluated at population level by comparison with pre-determined reference limits or cutoff points.

Indicators

- When indices jointly come with cutoff points they become **Indicators**.

The following are some of the examples of Indices & Indicators.

<u>Indices:</u> ➤ Weight-for-height (W/H): (Wasting), ➤ Height/Length-for-age (H/A) or (L/A): (Stunting), ➤ Weight-for-age (W/A): (Underweight), ➤ Body Mass Index (BMI), ➤ Mid Upper Arm Circumference (MUAC), ➤ Birth Weight (BW). ➤ Hemoglobin, clinical signs, serum retinol, urinary iodine etc.	<u>Indicators:</u> ➤ W/H Z-scores or H/A Z-scores or W/A Z-scores < -2SD, ➤ BMI < 18.5 kg/m², ➤ MUAC < 11.5 cm, BW < 2500g. ➤ Serum retinol levels < 0.7 μmol/L in children: Vitamin A deficiency.
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Anthropometric indicators for children less than 5 years of age

Indicator	What it measures/What it is used for
Low weight-for-height	WASTING (acute malnutrition). Wasted children are too light for their height (very thin). Wasting is the result of recent rapid weight loss or a failure to gain weight. Wasting can be reversed when conditions improve.
Low height-for-age or Low length-for-age	STUNTING (chronic malnutrition). Stunted children are too short for their age. Stunting develops over a long period as a result of inadequate dietary intake and/or repeated infections.
Low weight-for-age	UNDERWEIGHT (acute or chronic malnutrition, or both). Underweight children are too light for their age. Children may become underweight either because of wasting or stunting or both.

Dietary Indicators also include;

- ☑ Food intake or consumption surveys, which quantify the amounts and types of food eaten;
- ☑ Dietary Diversity Scores (DDS) and hunger scales;
- ☑ Food basket monitoring, which is usually carried out at food distribution sites or at household level in emergencies.

Indicators used for assessing micronutrient deficiencies (biochemical/laboratory & clinical indicators).

Indicators	What they measure	Contexts where used
- Clinical signs (pallor, tiredness, breathlessness and headaches) - Low haemoglobin	Iron deficiency anemia	STABLE SITUATIONS IDA results in decreased resistance to infection, impaired learning ability, low birth weight, decreased physical capacity and increased risk of death associated with pregnancy and childbirth. Clinical signs of anemia are monitored in surveillance systems in stable contexts. Women of child-bearing age, school children and children under five are most vulnerable to anemia and surveillance systems may focus only on these groups.
- Clinical signs (night blindness, bitot's spots, corneal xerosis, keratomalacia) - Low serum retinol	Vitamin A deficiency	EMERGENCIES AND STABLE SITUATIONS The consequences of VAD are tragic and include night blindness, irreversible blindness, growth retardation and increased susceptibility to infections and increased child mortality. Pregnant women are also prone to VAD and their children are likely to become deficient. Clinical signs are monitored in both stable contexts and emergencies. Children under five are usually monitored.
- Clinical signs (goitre and cretinism) - Low urinary iodine	Iodine deficiency	STABLE SITUATIONS Iodine deficiency results in disorders like goiter, impaired learning ability and reduced mental function (cretinism) and reproductive complications (miscarriages, still births and infant deaths). Clinical signs are monitored in stable contexts in areas where iodine deficiency is endemic, such as in mountainous areas.

1.8 The Design of Nutritional Assessment Systems

- The design of nutritional assessment systems is critical to use time and resources effectively. The assessment system used, the type & number of measurements selected, and the indices & indicators derived from these measurements will depend on a variety of factors including:

1.8.1 Factors influencing the design of a nutritional assessment systems

✓ Study objectives ✓ Sample size ✓ Sampling protocols ✓ Errors ✓ Accuracy ✓ Reproducibility or Precision	✓ Validity ✓ Confounding ✓ Sensitivity and specificity ✓ Prevalence ✓ Predictive value ✓ Time and resources
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1. **Study objectives:** can determine the type of nutritional assessment system to be used.

Examples of possible study objectives:

- Determine overall nutritional status of the population; *e.g.* EDHS
- Identifying areas, populations or subpopulations at risk of chronic malnutrition.
- Characterize extent and nature of malnutrition.
- Identify possible causes of malnutrition.
- Monitor changes in nutritional status.
- Evaluate efficacy or effectiveness of intervention programs.

2. **Sample size:** depends on the study objectives, the nature and scope of the study and expected outcomes.

3. **Sampling protocols** can be determined by sampling frame.

3.1 Probability sampling:

Mostly used when sampling frame is available. It involves random selection of subjects. It includes:

- ▶ Simple random sampling, Systematic random sampling, Stratified sampling, Multistage sampling & Cluster sampling.

3.2 **Non-probability sampling:** involves non-random selection of subjects. It is used in the absence of sampling frame. It includes;

- ▶ Consecutive sampling, convenience sampling (results in bias); quota sampling etc.

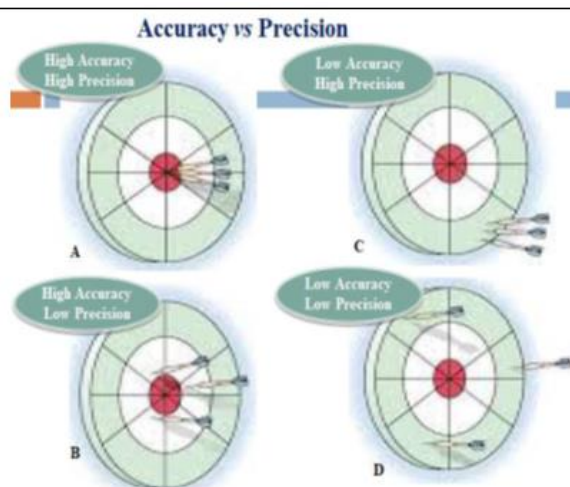
✚ **Consecutive sampling:** is the best of nonprobability sampling techniques: used in clinical research when it's feasible to recruit all available patients who fulfill the selection criteria.

✚ **Convenience sampling:** involves taking individuals in to a study when happened to be available during time of data collection.

4. **Accuracy:** is an extent to which a measurement/measured value/ is **correct** or **close** to “true or accepted value”.

5. **Precision or reproducibility (reliability):** the degree to which repeated measurements of the same variable; give the same (consistent) value, even if they are far from the accepted value. It can be assured by training; standardization; calibrated equipment; duplicated measurements.

N.B. A measurement can be reproducible but at the same time inaccurate.



6. Errors: two types of errors may occur during nutritional assessment.

6.1 Random errors: generate a deviation from the correct result due to chance alone or within subject variations. They reduce the **precision** of measurements.

6.2 Systematic errors: may arise in any nutritional assessment method and may become biased (a result departs from the true value). It reduces the **accuracy** of a measurement, but do not alter the **reproducibility** or **precision** of a measurement.

7. Validity: is an adequacy with which any measurement, test, index, or indicator reflects the parameter of interest.

8. Confounding: is a special type of bias that affects the outcome and validity of the study. For example cigarette smoking confounds the apparent relationship between coffee consumption and coronary heart disease (CHDs).

9. Prevalence: refers to the number of persons with disease in a given time period.

10. Predictive value: is the likelihood that a test correctly predicts the presence or absence of disease/ malnutrition.

11. Sensitivity: is the ability of index or indicator to identify & classify those persons who are genuinely malnourished.

- ▶ An index or indicator with 100% sensitivity correctly identifies all those individuals who are genuinely malnourished or no malnourished persons are classified as “well” (*i.e.*, there are no false negatives).

12. Specificity: is the ability of index or indicator to identify & classify those persons who are genuinely well nourished.

- ▶ If an index or indicator has 100% specificity, all genuinely well-nourished individuals will be correctly identified or no well-nourished individuals will be classified as “ill” (*i.e.*, there are no false positives).

1.8 Evaluation of Nutritional Assessment Indices

In Population Studies:

- Evaluated by comparison with distribution of reference values using percentiles, standard deviation scores (z-scores) and percent of median.

For Individuals:

- Values for nutritional assessment indices can be compared with either percentiles from reference populations or cutoff points.

1.9 Reference Distributions and Reference Limits

Reference distributions are also refers to reference values. They are preferably derived from a reference population (reference sample group) of *healthy persons*. They used to drive the reference limits and reference interval.

Examples of Reference Limits of Anthropometric indices in industrialized countries:

- ▶ 3rd or 5th percentile as unusually low.
- ▶ 95th or 97th percentile as unusually high.

WHO reference limits for hemoglobin*

Age	Hemoglobin g/L
0.5-5 y	<110
5-11 y	<115
12-13y	<120
Men	<130
Women	<120
Pregnant women	<110

Based on 2.5th percentiles
From Stolzhus & Dreyfuss (1998)

1.10 Cutoff points

Cutoff points are the relationship between nutritional assessment indices and subjects with low body stores, functional impairment, or clinical signs of deficiency. Their use is less frequent than reference limits because information relating tests or clinical sign of deficiency is often not available.

Sometimes more than one cutoff point can be selected.

- ▶ For example for BMI, two cutoffs of BMI are used for over nutrition (BMI>25 overweight, and BMI>30 obesity).
- ▶ Whereas three degrees of cutoffs are often used to define underweight in adults (mild, moderate & severe).

END OF CHAPTER ONE